

About Me...



Trailblazer
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- CTO & Managing Architect, Maple Digital
- Salesforce Certified Technical Architect (CTA)
- Salesforce MVP & MuleSoft Ambassador
- Architect Ambassador & Marketing Champion
- Founder ScaleUp Archs & Architect Ohana slack
- Boston Architect User Group Co-Leader
- Organizer for Northeast Dreamin & Co-Founder Architect Dreamin
- Advisory Board for Dreamin in Data



Salesforce Architecture

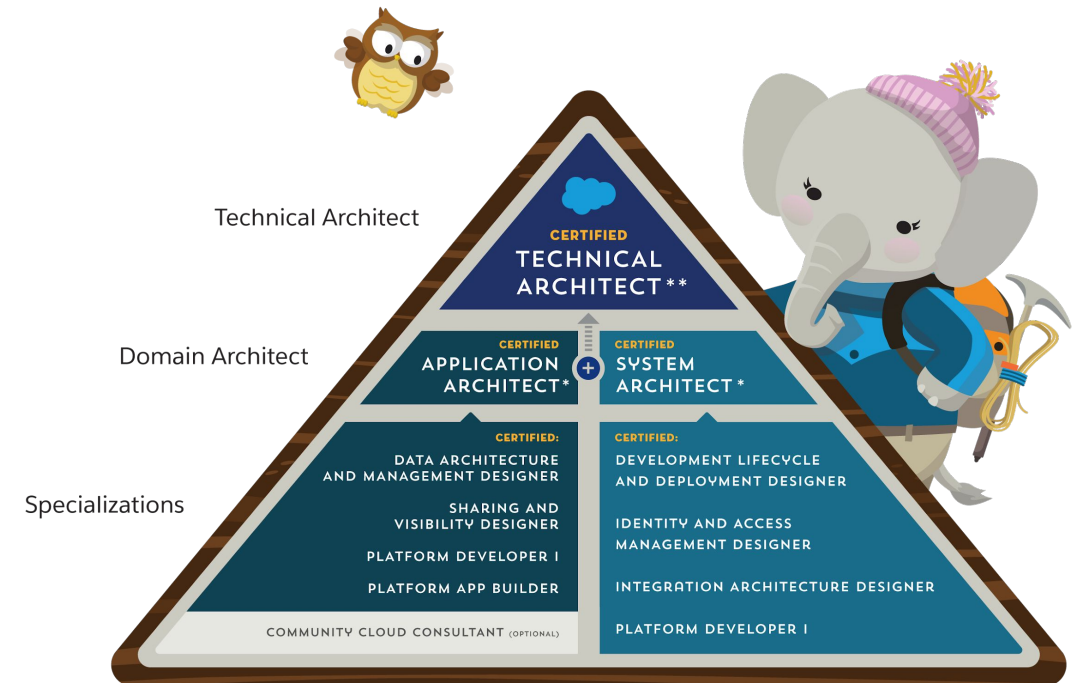
The buzzword

Questions that many of you might have:

- What is architecture all about?
- What exactly is an architect?
- What does an architect do?
- Is this something that I can do?
- Can I apply architect skills in my everyday job?
- If I don't consider myself an architect now, how can I start preparing myself to become one?



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Related Certification: Certified Administrator (optional)

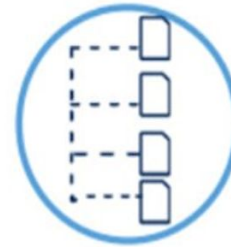
* Credential earned upon completion of exams within the specializations tier, no additional exam requirements.
** Credential earned upon successful completion of the Domain Architect tier and Review Board.

Why Architecture is Important

- Provides the building blocks
- Foundation for implementation
- Scalability & Performance
- System limitations
- Complex implementations
- Governance
- Regulations (GDPR, CCPA)
- Project success
- Reduce future costs
- Increase ROI



Large Data Volume



Large number of transactions



Complexity of transactions



Complexity of org



Data Skew

Traits of an Architect



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- ★ Has a thirst for knowledge
- ★ Can effectively communicate
- ★ Is curious and likes to do R&D to see how things will work
- ★ Has experience
- ★ Shows confidence
- ★ Has Leadership skills
- ★ Critical thinker
- ★ Lifelong learner
- ★ Future-thinking

Salesforce Architect

- ❖ Gains extensive platform knowledge
- ❖ Knows when to use platform features vs going custom
- ❖ Understands basic integration patterns
- ❖ Knows how to handle data on-platform
- ❖ Understands platform security features

CTA

- ❖ + enterprise knowledge
- ❖ + what will be lowest risk for project and bring highest ROI
- ❖ + which pattern to use when and why and how to secure data between all systems
- ❖ + what data should really be kept on-platform and when to move it off-platform
- ❖ + best security mechanisms to use for each system in the landscape

- ❖ Gain Salesforce platform knowledge
- ❖ Gain knowledge of common enterprise systems
 - What business functions they perform
 - What data they hold
- ❖ Know some common AppExchange apps for document management docgen,
- ❖ Understand integration patterns & tools to share data between systems
- ❖ Be able to answer questions and objections precise & concise
- ❖ Be able to justify decisions and communicate value
- ❖ Think about ROI of decisions
- ❖ Build up mental decision trees based on knowledge to know what option to choose based on business requirements as inputs
- ❖ Understand risks and how to mitigate them properly
- ❖ Understand how to mitigate issues with Large Data Volumes - on an off-platform
- ❖ Technical Knowledge + Business Acumen + Communication skills
- ❖ Be precise and concise, understand the keywords to use
 - Use language that can be easily understood by the business
- ❖ Taking the lead and proposing a well argued solution, not options



Think Big Picture

Will this scale beyond the current problem I am solving?

Things that can be taken into consideration when scaling your system:

- What kind of account model should be used (person vs account/contact or a mix)
- Record ownership in the role hierarchy - who should see what data? Will anyone own too many records? (ownership skew)
- Large Data Volumes - are you dealing with millions of records?
- Integrations with external systems
 - Point-to-point vs using middleware
 - Real-time callouts from Lightning Components
 - Is a response required? Does an update need to happen from an external system? How many API calls would be made?
- Licenses that are being used - are they correct for users access to objects?

Understand Solution Options



- ❖ Think about options but recommend the optimal solution
- ❖ Recommend simple solutions to solve complex problems
- ❖ One problem can have multiple solutions
- ❖ Understand considerations of your choice
- ❖ Be able to defend, justify, communicate effectively
- ❖ Understand trade-offs between other choices - why would you recommend one vs the other?
 - Understand the pros and cons of each and be able to recommend the best solution
- ❖ Understand the scalability implications of your choice - will it scale enough?
- ❖ Use knowledge and experience to make decisions



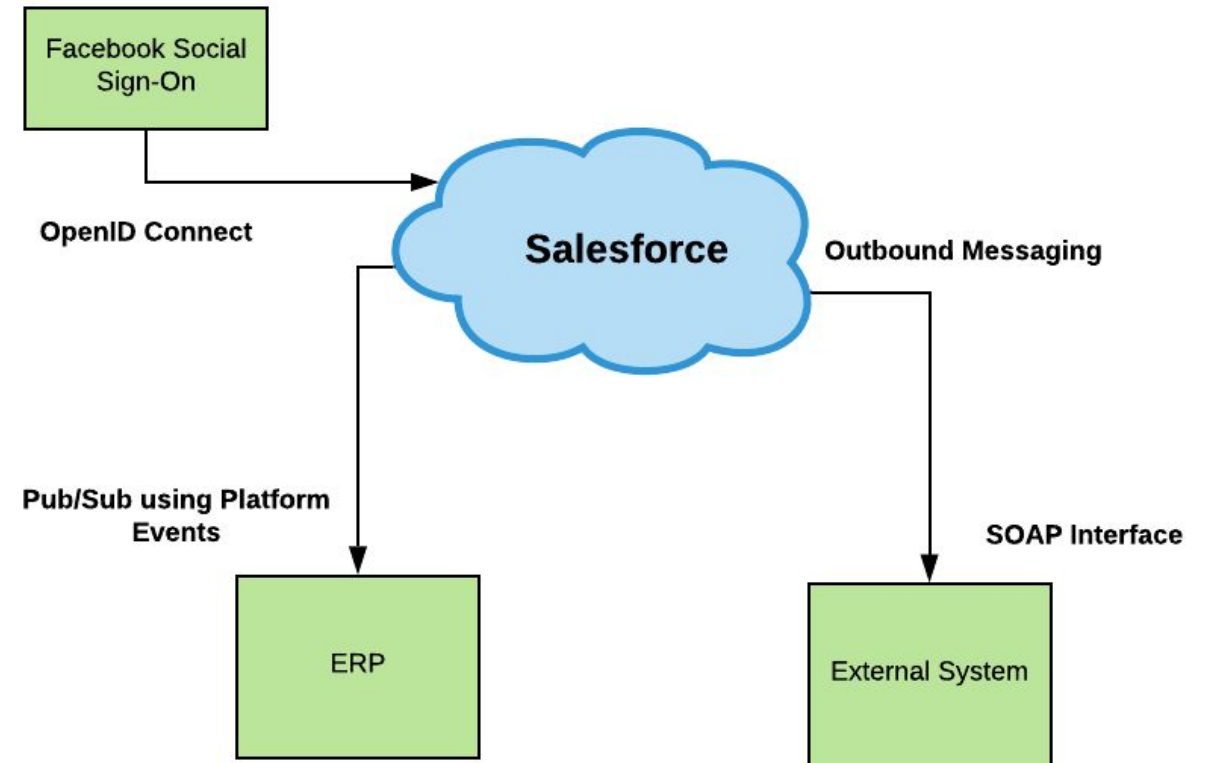
Create Diagrams

Creating a diagram can be as simple as showing the data model for some new objects that need to be created for a new business process.

Or maybe you have a new integration coming into the picture. How will this integration work? Will it be real-time? Asynchronous? What kind of authentication will be used? Draw it out!



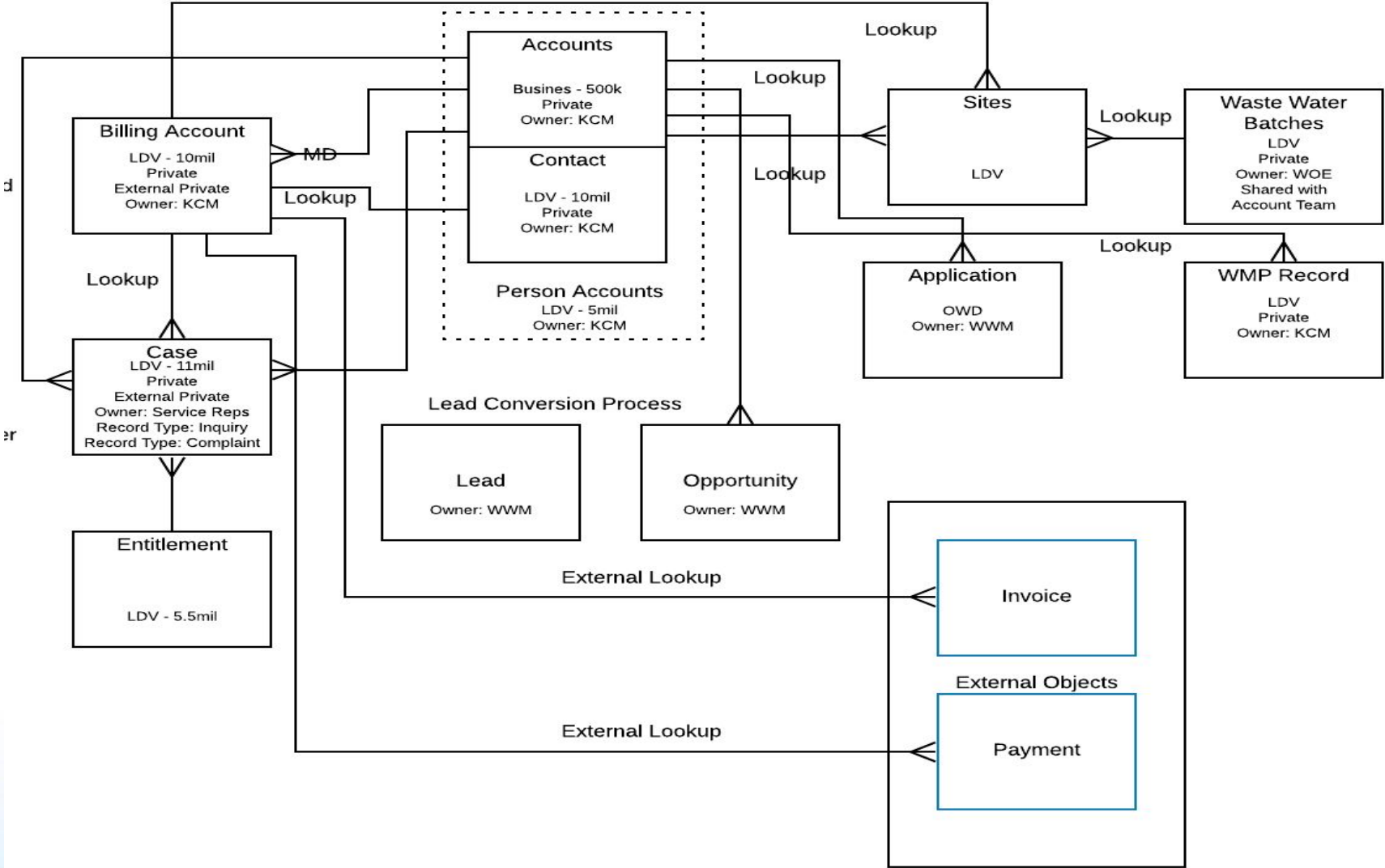
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Data Models/ERD



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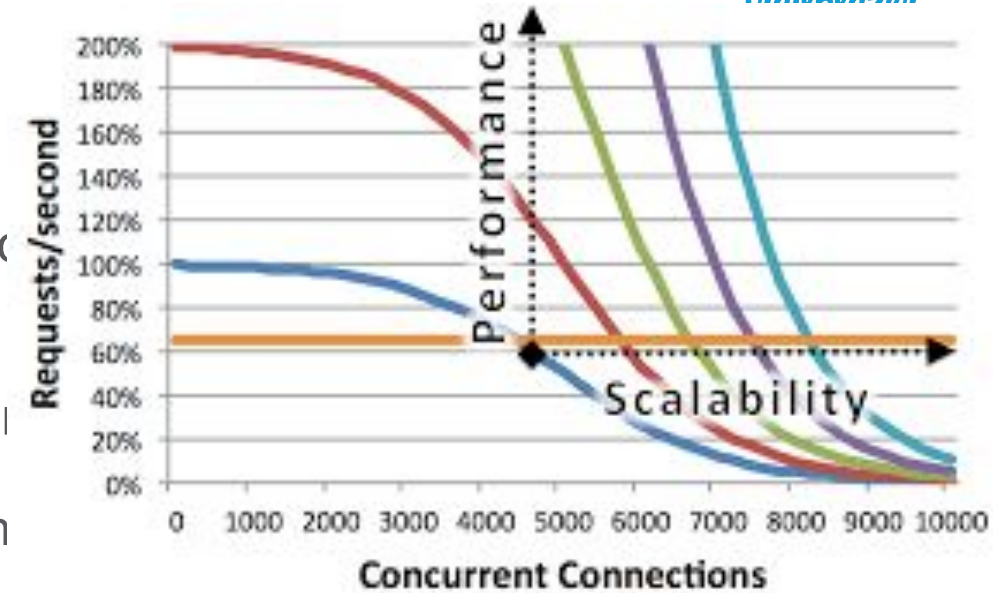


Understand Performance, Scalability & Security



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- ★ How to secure integrations
 - Ways to implement OAuth
 - JWT for Server-to-Server
- ★ One-way and Two-way TLS (MTLS), where and when you would want to use it
- ★ Data privacy
- ★ Data retention of on-platform data when LDV is a concern and how it can affect performance
 - Archival process to help mitigate your LDV - how would that work? Where will the data go?
- ★ How different integration patterns can affect performance
- ★ Data Visibility & Sharing - what data can users see?



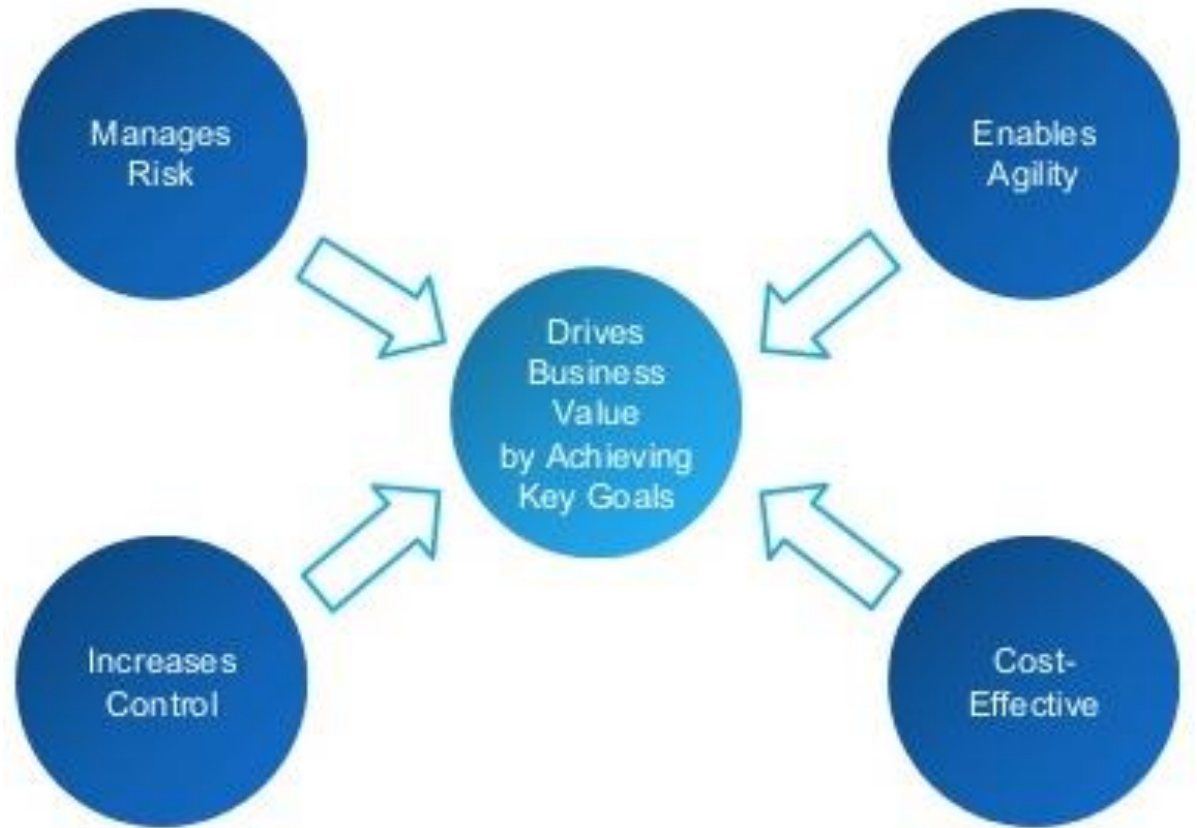
Follow Best Practices

Establish best practices for you and your colleagues to follow. Can you identify areas of improvement?

- Coding practices - are they being followed?
 - Implement code analysis tools such as PMD, ESLint, CodeScan
- Is there a mature devops process?
 - Implement devops automation
 - Suggest the use of tools such as Github, CI/CD, SFDX, Copado
- Is data quality being maintained?
 - Use of validation rules
 - Use of Duplicate rules
- Are you relying on automation tool when necessary? (Clicks vs Code)
 - Don't over-customize

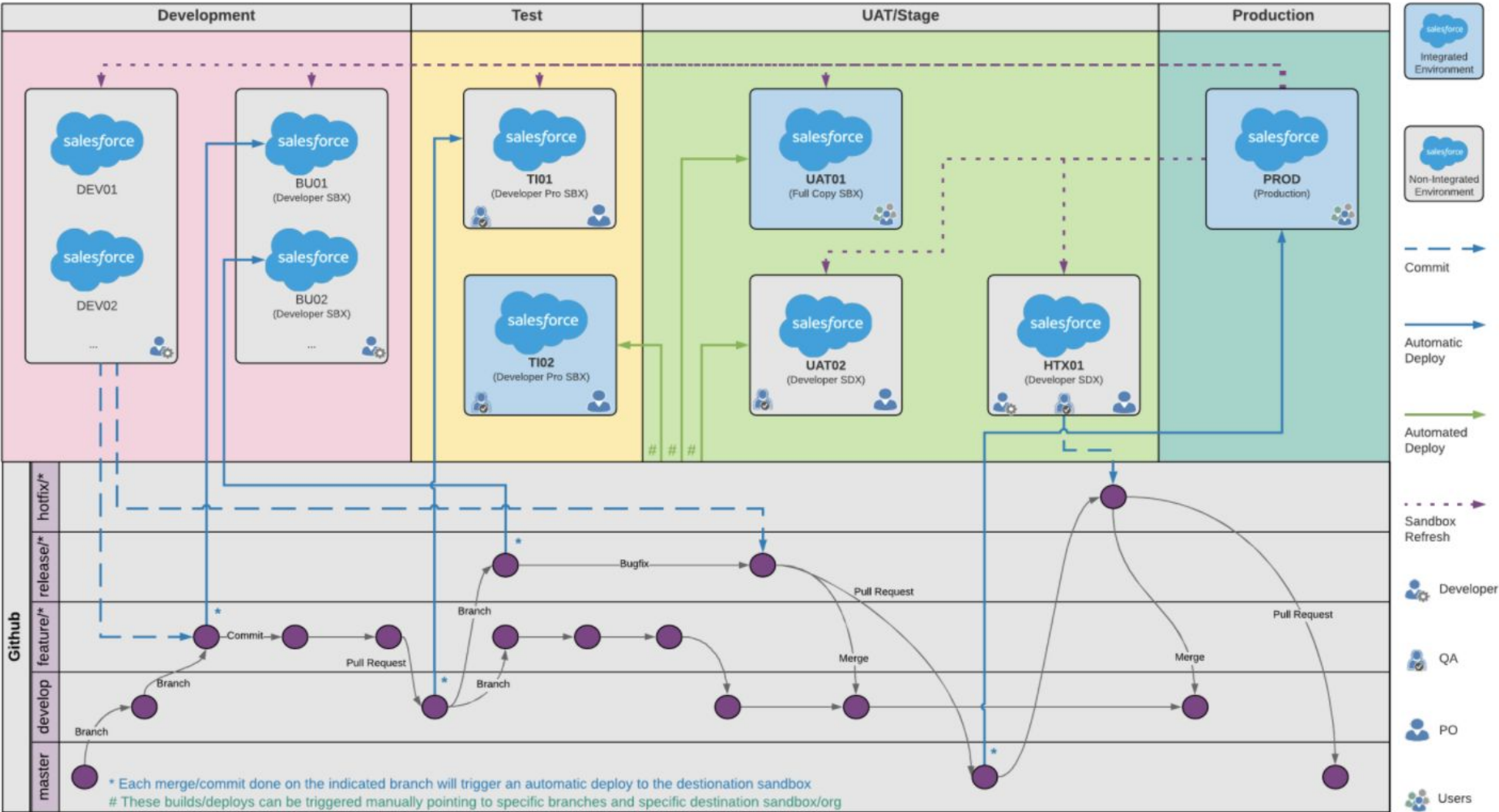
Establish Governance

- Have a defined release management process and assign someone to oversee it
 - Define a daily developer workflow
- Set up a Center of Excellence
- Set up an Architecture Review Board
 - Review change requests before they get into the backlog
- Follow a methodology such as Agile or a hybrid such as Wagile/Scrumfall
- Define a testing strategy



Identify use of Environments

Make sure the right environments are being used



Understand Your Org's Security



- When changes need to be made you will be better prepared
- When things like record locking happen you can contribute to solving the problem
- Look at things such as:
 - Role Hierarchy
 - Number of roles - max 500k (with approval from Salesforce)
 - Can have an impact on sharing recalculation causing performance issues and record locks
 - Sharing Rules - number of sharing rules, 50 CB per object, max 500 total
 - Too many sharing rules can slow down inserts and updates
 - Org-wide Defaults - are they providing enough security?
 - Profiles and Permission Sets
 - Are there too many profiles? Should some permissions be migrated to permission sets?



Understand Capabilities of the Clouds

There may be features in another Salesforce cloud that are available to solve a problem

- Learn about the capabilities of each of the clouds and how you might be able to incorporate them
 - Sales Cloud Features
 - Use of Leads, Lead assignment, web-to-lead
 - Service Cloud Features
 - Digital Engagement
 - Omni-Channel, Skills-based Routing
 - Community Cloud
 - External sharing
 - Marketing Cloud
 - Email marketing, customer journeys



Engage with Stakeholders



- Communicate with stakeholders some of your ideas
 - Project Management, Project Sponsors, Product Owners
- Help fill in any missing gaps there may be in the requirements
- Help to communicate project objectives and initiatives
- If you want to recommend something, be ready to defend and justify your selections



Understand Value for the Business



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- ❖ Maintainability - clicks vs code
- ❖ Simple OOTB solutions can sometimes suffice - lower cost, lower risk
- ❖ Solutions that bring higher ROI are more desirable
- ❖ Avoid solutions that will create technical debt
- ❖ Total cost of ownership
 - Going with an appexchange app over building custom to solve a business problem
- ❖ Time to market - how fast can you get the functionality to your end users
- ❖ Does it actually make sense to do the work or can a process be changed to fit what is already OOTB?
- ❖ Licensing costs - Platform, Service vs Sales or both?



Understand User Adoption



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- ★ Understand how users are actually going to use the solution/system
- ★ Think about driving user adoption through your solution
- ★ How many clicks does it take for a user to perform a task?
- ★ Can a Screen Flow help with a complex process?
- ★ Will your integrations have impacts on User Experience? Choosing Request & Reply vs Fire & Forget
- ★ UI/UX is an important aspect to think about when designing solutions
- ★ Process Mapping more holistic e-2-e solutions and user experience



Become a Trusted Advisor



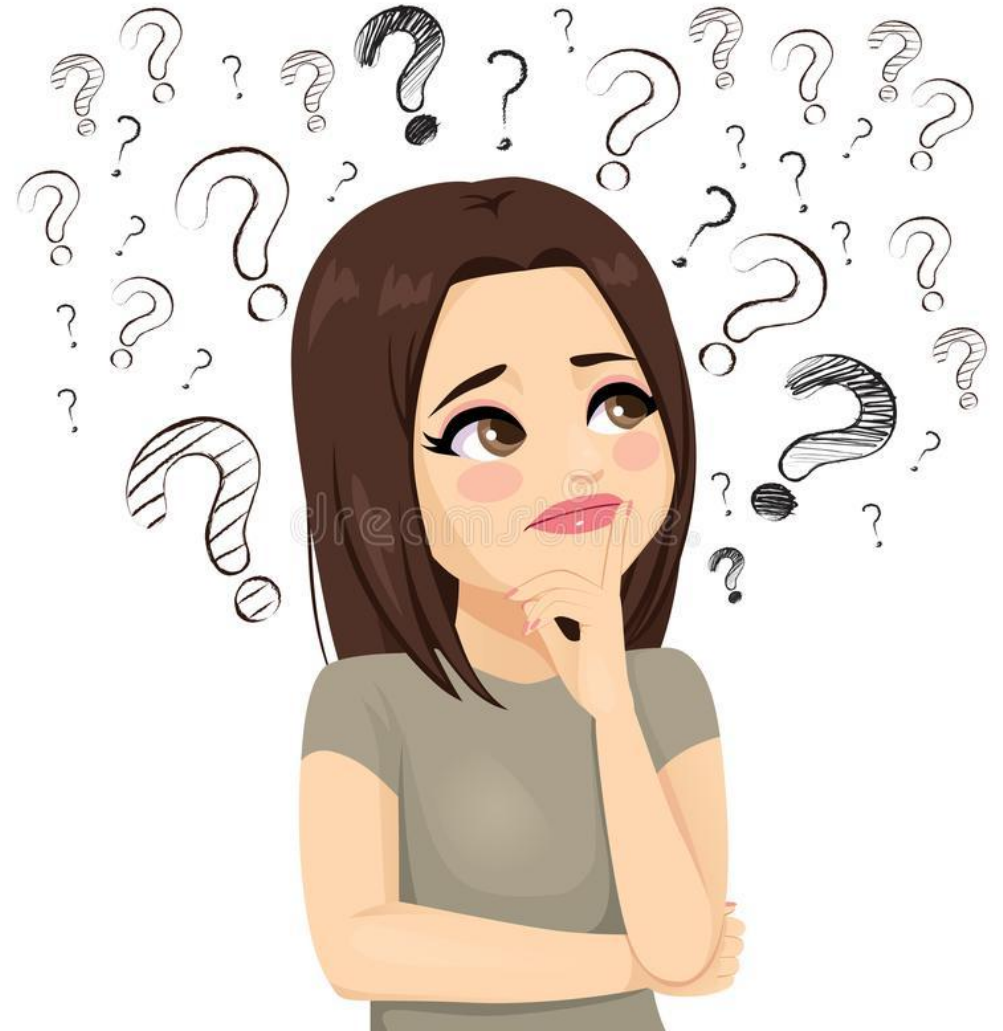
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- Advise on process best practices -
 - DevOps, governance, project management
- Guidance, mentoring, help companies implement process
- Be able to explain benefit to the business
 - How will it be less risky than another option
 - How it will benefit the project timeline
- Look at big picture, future thinking - 3, 4, 5 years down the road not just today -
 - How does it scale with growth
- Engage with stakeholders
 - How does something solve a business problem
- Ownership and the role of trusted advisor are what differentiates an architect from a consultant/dev/admin



Be Curious!

- What's really going on behind the scenes?
 - Ask questions about the technical aspects of how a solution is implemented
- Poke around in your org and identify areas for improvement
- Play around in a sandbox doing R&D
 - How to share data with internal and external users
 - Can customers be added to case teams? Can they see the case?
 - How can you display data securely from an external object to an external user?



Recommended Reading



<u>Book Title</u>	<u>Recommendation</u>
Becoming a Salesforce Certified Technical Architect	Must read - should be part of any CTA journey
Salesforce Data Architecture and Management	Strongly recommend
Enterprise Integration Patterns	Strongly recommend
Developing Data Migrations and Integrations with Salesforce: Patterns and Best Practices	Strongly recommend
Salesforce B2C Solution Architect's Handbook	Strongly Recommend for B2C Solution Architect exam
Solving Identity Management in Modern Applications: Demystifying OAuth 2.0, OpenID Connect, and SAML 2.0	Recommended, especially if you struggle with Identity concepts
Salesforce Architect's Handbook: A Comprehensive End-to-End Solutions Guide	Nice to have
Salesforce Lightning Platform Enterprise Architecture: 3rd Edition	Nice to have
Mastering Salesforce DevOps: A Practical Guide to Building Trust While Delivering Innovation	Nice to have
The Practice of Enterprise Architecture: A Modern Approach to Business and IT Alignment	Nice to have
Certified Enterprise Architect All-in-One Exam Guide	Nice to have

Architect Resources



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- Architect Ohana Slack (and YouTube): <https://bit.ly/3OfZwm4>
- Architect Ohana **YouTube:**
<https://www.youtube.com/channel/UCfWf8b7Dyitl0rQf7oswWIA>
- Benelux CTA Study Group: <https://www.meetup.com/CTA-Study-Group>
- Salesforce Architects: <https://architect.salesforce.com/>
- Ladies Be Architects: <https://ladies-be-architects.com/>
- Salesforce Architect Community: <https://sforce.co/2wWgdAs>
- Apex Hours: <https://www.apexhours.com/>
- Salesforce Architects on Medium: <https://medium.com/salesforce-architects>
- Gemma Blezard: <http://gemmablezard.com/>
- Susannah Kate Plaisted: <https://www.accidentaladmintoawesomearch.com/>
- CTA Gang of Four: <https://ctagof.com/>

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